

33. (g) Implementation Plan: B.Tech. in Electronics and Communication Engineering with Specialization in Data Science

Semester – I						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH101T	Communicative English	2	1	0	3	
21MAB101T	Calculus and Linear Algebra	3	1	0	4	
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5	
21MES102L ¹	Engineering Graphics and Design	0	0	4	2	
21EES101T	Electrical and Electronics Engineering	3	1	0	4	
21CYM101T ¹	Environmental Science	1	0	0	0	
21PDM101L ¹	Professional Skills and Practices	0	0	2	0	
21LEM101T ¹	Constitution of India	1	0	0	0	
Total Credits 18						
Semester – III						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4	
21PDH209T ¹	Social Engineering	2	0	0	2	
21CSS201T	Computer Organization and Architecture	3	1	0	4	
21ECC201T ²	Solid State Devices	3	0	0	3	
21ECC203T	Digital Logic Design	3	0	0	3	
21ECC205T	Electromagnetic Theory and Interference	3	0	0	3	
21ECC211L ¹	Devices and Digital IC Laboratory	0	0	4	2	
21LEM201T ¹	Professional Ethics	1	0	0	0	
21PDM201L ¹	Verbal Reasoning	0	0	2	0	
21LEM202T ¹	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3	
Total Credits 24						
Semester – V						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB302T	Discrete Mathematics	3	1	0	4	
21ECC301P ¹	Microprocessor, Microcontroller, and Interfacing Techniques	3	1	0	4	
21ECC303T ²	VLSI Design and Technology	3	0	0	3	
21ECC311L ¹	VLSI Design Laboratory	0	0	4	2	
E	Professional Elective – II				3	
O	Open Elective – I				3	
21GNP301L ¹	Community Connect	0	0	2	1	
21PDM301L ¹	Analytical and Logical Thinking Skills	0	0	2	0	
21LEM301T ¹	Indian Art Form	1	0	0	0	
Total Credits 20						
Semester – VII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21GNH401T	Behavioral Psychology	2	1	0	3	
21ECC401T ²	Wireless Communication and Antenna Systems	3	0	0	3	
21ECC402P ¹	Computer Communication and Network Security	2	1	0	3	
E	Professional Elective – V				3	
E	Professional Elective – VI				3	
O	Open Elective –III				3	
Total Credits 18						

Semester – II						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH102T	Chinese	2	1	0	3	
21LEH103T	French					
21LEH104T	German					
21LEH105T	Japanese					
21LEH106T	Korean					
21LEH107T	Spanish					
21LEH108T	Russian					
21GNH101J	Philosophy of Engineering	1	0	2	2	
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4	
21CYB101J	Chemistry	3	1	2	5	
21ECC101J	Electronic System and PCB Design	2	0	2	3	
21CSS101J	Programming for Problem Solving	3	0	2	4	
21BTB103T	Biology	2	0	0	2	
21MES101L ¹	Basic Civil and Mechanical Workshop	0	0	4	2	
21PDM102L ¹	General Aptitude	0	0	2	0	
21GNM101L ¹	Physical and Mental Health using Yoga	0	0	2	0	
21GNM102L ¹	National Service Scheme					
21GNM103L ¹	National Cadet Corps					
21GNM104L ¹	National Sports Organization					
Total Credits 25						
Semester – IV						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB203T	Probability and Stochastic Processes	3	1	0	4	
21ECC202T	Analog and Linear Electronic Circuits	3	0	0	3	
21ECC204T ²	Signal Processing	3	0	0	3	
21ECC222L ¹	Analog and Linear Electronic Circuits Laboratory	0	0	4	2	
21CSC206T	Artificial Intelligence	2	1	0	3	
E	Professional Elective-I				3	
21DCS201P ¹	Design Thinking and Methodology	1	2	0	3	
21PDM202L ¹	Critical and Creative Thinking Skills	0	0	2	0	
Total Credits 21						
Semester – VI						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21CSS303T	Data Science	2	0	0	2	
21ECC302T	Analog and Digital Communication	3	0	0	3	
21ECC304T ²	Microwave and Optical Communication	3	0	0	3	
21ECC322L ¹	Communication Laboratory	0	0	4	2	
E	Professional Elective – III				3	
E	Professional Elective – IV				3	
O	Open Elective – II				3	
21ECP302L ¹	Project	0	0	6	3	
21ECP303T ¹	MOOC	3	0	0		
21PDM302L ¹	Employability Skills and Practices	0	0	2	0	
21LEM302T ¹	Indian Traditional Knowledge	1	0	0	0	
Total Credits 22						
Semester - VIII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21ECP401L	Major Project	0	0	30	15	
21ECP402L	Major Project	0	0	20	10	
21ECP403L	Internship#	0	0	10	5	
Total Credits 15						

#Students have to register either 21ECP401L or 21ECP402L and 21ECP403L both in eighth semester